What Makes A Good App? - Analyzing User Ratings and UX Patterns in the Apple Store

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Abstract

This project investigates the relationship between app characteristics and user ratings in the **Apple Store**. The goal is to uncover data-driven insights relevant to **UI/UX** design. Using a data set of more than 1.2 million applications collected in October 2021, we applied exploratory data analysis and statistical methods to examine how factors such as app category, pricing model, app size, and review volume relate to user satisfaction. After cleaning and reprocessing the data set to 184,544 applications, five key visualizations were produced to surface patterns. Our analysis reveals that the majority of apps between 4.0 and 5.0 stars, with a mean rating of 4.13, reflecting Apple's strict design standards

Code: `https://github.com/SamyPrabhak/Intro_to_DataSci_FinalProject`

Video: `https://drive.google.com/file/d/1omxkeVBIDvrz9YtmGzFN5LUYUmZPJX5t/view?usp=sharing`

1 Introduction

Mobile applications have become one of the most visible and widely used products of UI/UX design. With more than 1.2 million apps available on the Apple App Store (data up to October 2021), users are faced with an overwhelming number of choices and their ratings serve as one of the most direct, large-scale measures of user satisfaction available.

Apple is known for enforcing strict Human Interface Guidelines (HIG) for all submitted apps, meaning that a baseline level of design quality is required just to enter the marketplace. Despite this high bar, user ratings still vary enormously between apps and categories, which suggests that factors beyond baseline compliance drive user satisfaction.

We have a personal interest in UI/UX design and a desire to understand user satisfaction not just qualitatively, but empirically. By applying data science techniques to real App Store data, we aim to answer the following research question:

What app characteristics are most strongly associated with higher user ratings on the Apple App Store?

We examine the role of app category, pricing model (free vs. paid), app size, and review volume in predicting user satisfaction. Our goal is to surface actionable insights that can inform design decisions that bridging the gap between data science and UX practice.

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

2 Related Work

Research at the intersection of data science and mobile app store analysis has grown significantly over the past decade, providing important context for this project.

2.1 App Store Mining

Harman, Jia, and Zhang (1) were among the first to formally study app stores as software repositories, performing correlation analysis on price, downloads, and ratings. This work provides a methodological foundation for our analysis, which extends their approach to a larger Apple App Store dataset of 1.2 million apps.

2.2 Data Science on App Store Ratings

Vo et al. (2) conducted a closely related study using EDA and clustering techniques on Apple App Store data sourced from Kaggle, the same platform used in this project. The analysis of numeric app attributes to classify apps by rating directly parallels our research approach, and their findings on rating distributions across categories informed our category-level analysis.

2.3 Ratings and Review Volume

Sällberg, Wang, and Numminen (3) tracked 341 gaming and productivity apps from their App Store release, finding that rating volume and average score significantly influence app downloads. This supports our survival bias finding, apps with high review volumes tend to maintain higher ratings over time.

These studies confirm the methodology used in this paper and provide a solid basis for data-driven app store analysis. This project expands on earlier research that looked at app store mining, clustering techniques, and the impact of review volume on app performance by using exploratory data analysis on a sizable dataset of 1.2 million apps from the Apple App Store. Our project's results have both validated and refuted previous research, demonstrating that, in contrast to popular belief, the price model has no appreciable effect on user satisfaction and supporting the relationship between review volume and app quality.

3 Method

3.1 Dataset

The primary dataset we used is the Apple App Store Apps dataset, where it is publicly available on Kaggle (4) It was Collected in October 2021 via Python and Scrapy. It contains over 1.2 million apps records across 21 attributes, making it one of the largest publicly available App Store datasets. The dataset will be downloaded directly from Kaggle as a CSV file and loaded into a Jupyter Notebook using Pandas.

3.2 Data Exploration

The dataset required the following steps before analysis:

- **Missing values:** Rows with missing or zero `Average_User_Rating` values were dropped, as rating is the primary outcome variable.
- **Size conversion:** The `Size_Bytes` column was converted to megabytes (MB) for more interpretable analysis.
- **Free vs. Paid classification:** A binary column was derived from the `Price` column (0.00 = Free, >0.00 = Paid).
- **Rating count filtering:** Apps with fewer than 10 reviews were excluded to avoid unreliable rating averages.
- **Duplicate removal:** Duplicate app entries were identified by `App_Name` and `Primary_Genre` and removed.

- **Date parsing:** `Released` and `Updated` columns were parsed using `pd.to_datetime()` to enable time-based analysis.

Then begin with univariate and bivariate exploratory data analysis (EDA) to understand distributions and surface initial patterns:

- Distribution of ratings across all apps (histogram)
- Average rating per category (horizontal bar chart)
- Distribution of ratings for free vs. paid apps (box plot)
- Correlation matrix across numeric features (heatmap)
- Review volume vs. average user rating (scatter plot)

3.3 Feature Engineering

New features were derived to enrich the analysis:

- Size (MB): App size converted from bytes to megabytes for more interpretable analysis.
- Free vs. Paid: A binary classification derived from the Price column (0.00 = Free, >0.00 = Paid).
- Cleaned Dates: Released and Updated columns parsed into datetime format to enable time-based analysis.

Note: **Days Since Update** and **Log Reviews** were initially planned but not implemented. The Days Since Update feature was excluded because it a static dataset collected in 2021 and does not reflect app update activity beyond that date, making recency calculations unreliable for current analysis.

3.4 Statistical Analysis

Descriptive statistics and group comparisons were applied to answer each research question:

- **Mean rating comparisons** across app categories using groupby-aggregation to identify highest and lowest rated genres
- **Free vs Paid comparison** using box plot visualization and descriptive statistics (mean, median, std) to evaluate pricing impact on ratings
- **Correlation analysis** using a heatmap to measure relationships between numeric features (rating, reviews, size, price)
- **Review volume analysis** using a scatter plot with log scale to examine the relationship between popularity and user satisfaction

3.5 Predictive Modeling

This phase was not implemented as correlation analysis showed no single numeric feature strongly predicts app ratings (all correlations < 0.10), making a regression model ineffective. Instead, our analysis led to the following data-driven predictions:

- **App category is the strongest differentiator**: Magazines and Newspapers consistently outperform Utilities and Business apps
- **Price does not predict quality**: Free and paid apps rate virtually identically (4.13)
- **Review volume predicts survival**: Apps with 10,000+ reviews converge toward 4.5+ ratings
- **The App Store acts as a natural UX filter**: High review counts are an indirect but reliable proxy for design quality

4 Evaluation

We analyze the patterns observed across the five visualizations and assess how well they support our research question regarding app characteristics and user ratings.

4.1 Findings

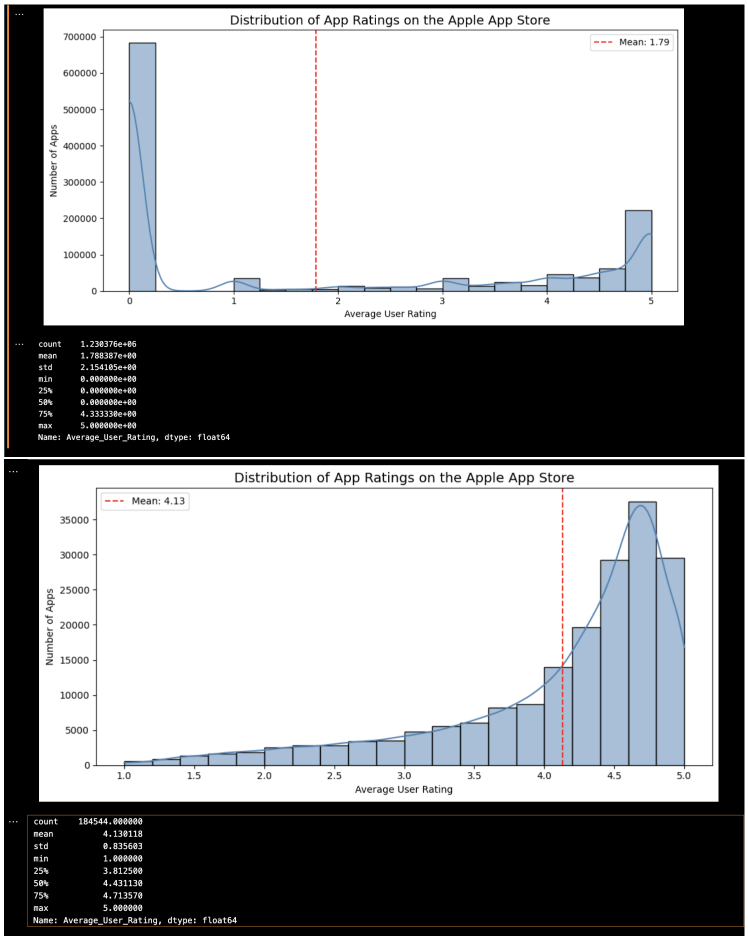

Figure 1: Comparison of ratings before and after cleaning.

Before Cleaning

- Count: 1,230,376 apps
- Mean: 1.79 (dragged down by unrated apps with 0)
- Huge spike at 0 — most apps had no ratings

After Cleaning

- Count: 184,544 apps
- Mean: 4.13 (accurate, reflects real user ratings) Min: 1.0, Max: 5.0, Median: 4.43
- Clean distribution clustering between 4.0 – 5.0

From Figure 1, the cleaning process removed over 1 million unrated or low-review apps, revealing the true distribution of user satisfaction on the Apple App Store. This before/after comparison shows why data-preprocessing is a critical step in any data science analysis as raw data can be extremely misleading without proper cleaning.

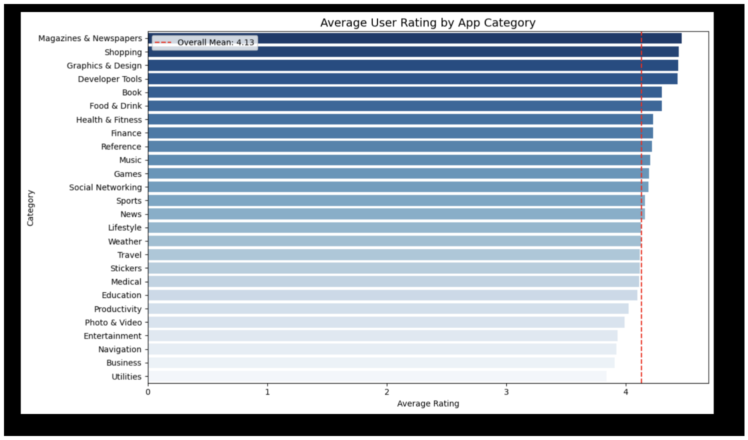

Figure 2: Horizontal Bar chart of Average User Rating by App Category.

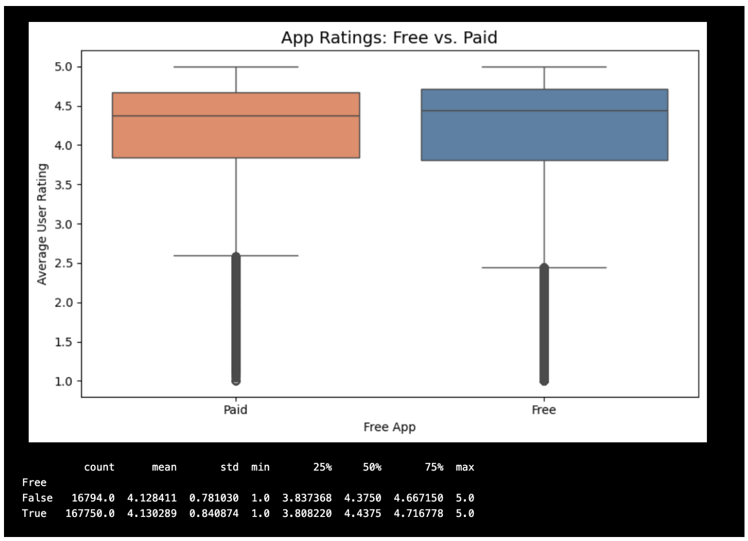

Figure 3: Boxplot of App Ratings free .

The horizontal bar chart (Figure 2) reveals significant variation in average user ratings across app categories. The **Magazines and Newspapers** ranked highest, followed by **Shopping and Graphics and Design**, while **Utilities, Business, and Navigation** ranked lowest. The red dashed line represents the overall mean of 4.13, showing which categories perform above and below average. Therefore,the finding suggests that niche, content-focused apps consistently outperform broad utility-based apps in user satisfaction.These apps help users, as specialized apps have clearer expectations and are more easily satisfied by well-designed interfaces.

The box plot (Figure 3) compares the distribution of ratings between free and paid apps. Both groups show nearly identical medians. This finding directly contradicts our initial hypothesis that paid apps would rate higher on the Apple App Store with their strict Human Interface Guidelines. It create a quality baseline that levels the playing field between free and paid apps, suggesting that pricing alone is not a reliable signal of UX quality.

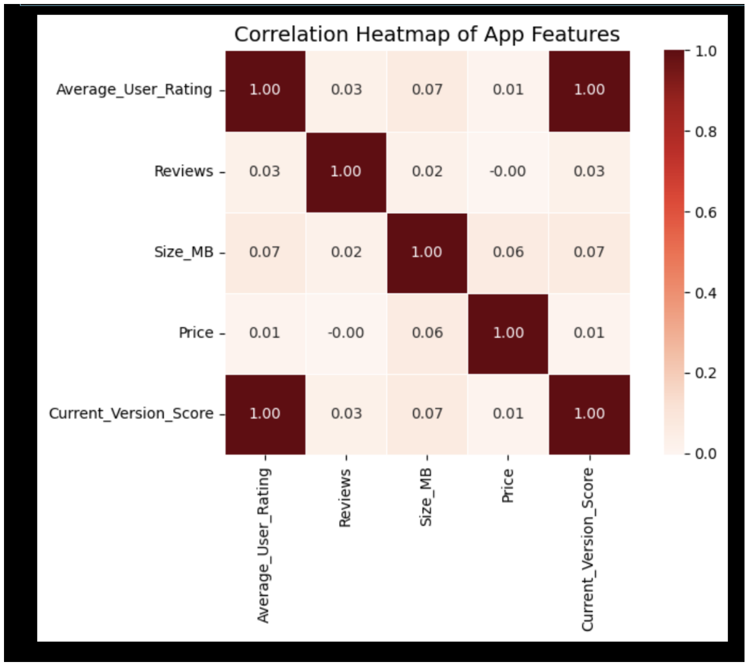

Figure 4: A Correlation Heatmap of App Features.

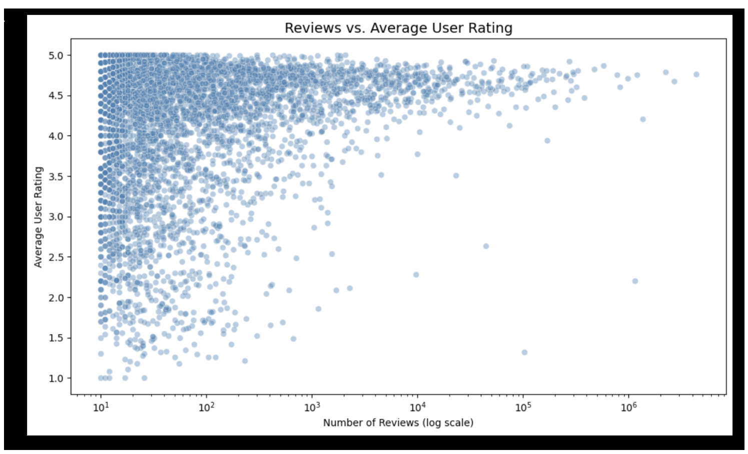

Figure 5: A Scatterplot between Reviews vs Average User Rating.

The heatmap (Figure 4) shows correlations between numeric app features. All the correlations with AverageUserRating were near zero: Reviews (0.03), Size MB (0.07), and Price (0.01). Meaning that none of these features strongly predict user ratings. This suggests that what drives user satisfaction in the Apple App Store goes beyond measurable numeric attributes and likely lies in factors that our dataset was unable to capture.

The scatter plot (Figure 5) reveals a clear survival bias effect in the Apple App Store. Apps with few reviews show ratings a wide range from 1.0 to 5.0, while apps with thousands of reviews consistently converge between 4.5 to 5.0. A log scale was used on the x-axis to handle the wide range of review counts.

From this we confirm that the App Store acts as a natural UX filter. Poorly designed apps get abandoned before accumulating reviews, while well-designed apps retain users and earn higher ratings over time.

4.2 Evaluation Table

Table 1: Research question coverage. Each question is addressed using a specific method, with key findings summarized.

Research Question	Method Used	Finding
Does category affect ratings?	Bar chart, groupby	Yes - Magazines highest, Utilities lowest
Does price affect ratings?	Box plot + descriptive stats	No - both $\sim$4.13
Does size correlate with ratings?	Correlation heatmap	No - correlation = 0.07
Does review volume affect ratings?	Scatter plot	Yes - survival bias confirmed

Table 2: The results of the hypothesis

Hypothesis	Result
Free apps rate lower than paid	False
Health and Fitness tops categories	False
App size correlates with rating	False
Popular apps rate higher	True

4.3 Coverage

- Started with **1,230,376** apps
- After data cleaning: **184,544** apps
- Removed: Low review apps, Duplicates, and Missing ratings

5 Discussion

Our analysis of Apple App Store(184,544) apps revealed several surprising and meaningful insights about user satisfaction.

The most unexpected finding was that free and paid apps rate virtually identically (4.13), directly contradicting our initial hypothesis, that paid apps would have higher ratings. This suggests that Apple's strict design guidelines create a quality floor that levels the playing field between free and paid apps.Hence,why price alone is not a signal of quality.

Category emerged as the strongest differentiator of user satisfaction. As seen in Figure 2, **Magazines and Newspapers** topped the ratings while **Utilities** ranked lowest. This aligns with UX research suggesting that niche, content-focused apps better meet user expectations because their purpose is clearly defined, users know exactly what they are getting and are able to access content easily due to good UI design.

The correlation heatmap revealed that no single numeric feature,price, size, or review count,strongly predicts ratings. This is a significant finding for UX designers: it suggests that measurable app metadata alone cannot explain user satisfaction. What drives ratings is likely qualitative-design polish, ease of use, and responsiveness,factors that are difficult to capture in structured data.

Connection to UI/UX Design -> These findings have direct implications for UI/UX designers:

- Investing in design quality over pricing strategy is what drives user satisfaction.
- Category-specific UX expectations matter — designers should study the conventions of their target category.
- Developer responsiveness and ongoing UX improvements are critical for long-term app survival.
- Data science can provide empirical evidence for design decisions that are often made based on intuition alone.

6 Limitation

- **Dataset is a static measure :** The dataset was collected in October 2021 and does not reflect apps released, updated, or removed after that point. The Current App Store trends may differ significantly.
- **Recent Updates is unmeasurable :** The dataset only captures data up to October 2021, the Days Since Update feature could not be reliably calculated for current analysis.
- **Low review apps excluded :** Apps with fewer than 10 reviews were removed during cleaning, which can ignore or underepresent newer or niche apps that have not yet accumulated sufficient user feedback.
- **Correlational findings only :** No strict conclusions can be concluded from this analysis.The High review volume may correlate with high ratings, but does not necessarily cause them.
- **Fake review noise :** App ratings may be subject to review manipulation through fake or incentivized reviews. This would introduce noise into the analysis.
- **Numeric features only :** The analysis is limited to structured numeric and categorical data.Qualitative factors such as design polish, ease of use, and visual aesthetics,which likely drives user satisfaction most,are not captured in the dataset.
- **Single platform :** Findings are specific to the Apple App Store and may not be generalized to other platforms such as the Google Play Store.

7 Conclusion and Future Works

In conclusion, this sets out to investigate what app characteristics are most strongly associated with higher user ratings on the Apple App Store, combining data science methods with a UI/UX design perspective. Analyzing 184,544 cleaned app records from a dataset of 1.2 million apps, our exploratory data analysis surfaced several meaningful findings.

Contrary to our initial hypotheses, neither price nor app size meaningfully predicted user ratings. Instead, app category emerged as the strongest differentiator, with **Magazines & Newspapers** consistently outperforming **Utilities** and **Business** apps. Most notably, the survival bias effect confirmed that the Apple App Store functions as a natural UX filter, poorly designed apps are abandoned before accumulating reviews, while high quality apps retain users and earn higher ratings over time.

These findings demonstrate that data science can provide empirical grounding for UX design decisions, moving beyond intuition toward population-level insights drawn from real user behavior.

7.1 Future Work

Building on these findings, future research could explore the following:

- **Sentiment analysis :** on user review text to capture qualitative UX factors not present in numeric data.
- **Cross-platform comparison:** with other app store applications like Google Play Store to test whether findings generalize across ecosystems.
- **More recent dataset :** A dataset post-2021 to reflect current App Store trends.It would be interesting to conduct analysis post-pandemic(after 2022) to see if the findings change or remain the same.
- **Machine learning models :** ML models could be trained on richer features including review text and UI screenshots. From this we could learn about good UI Practices and help us to analysis better.

References

[1] Mark Harman, Yue Jia, and Yuanyuan Zhang. App store mining and analysis: MSR for app stores. In *Proceedings of the 9th IEEE Working Conference on Mining Software Repositories (MSR)*, 2012. doi: `https://doi.org/10.1109/MSR.2012.6224306`.

[2] *Characteristics Classification of Mobile Apps on Apple Store Using Clustering*. Journal of Data Analysis and Information Processing, 2020. doi: `https://doi.org/10.4236/jdaip.2020.82004`.

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